

CRAFTLANEE

Connect • Collaboration • Create

E-LEARNING PLATFORM

Learning Management System (LMS) Portal

Project & Technology Documentation

September 2026

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1. Project Overview

The Craftlanee E-Learning Platform is a full-featured Learning Management System (LMS) built on Django. It brings together students, instructors, and administrators on a single portal that covers the complete learning lifecycle - course creation, enrollment, lesson delivery, quizzes, progress tracking, skill mapping, gamification, certification, in-app messaging, and an AI-powered mentor chatbot - wrapped in an age-adaptive theme (school / intermediate / college / university).

The platform is organised as a set of focused Django apps, each owning one part of the domain (accounts, courses, students, instructors, adminpanel, mentor_ai, career, messaging, utils), communicating through a shared MySQL database and rendered through server-side Django templates.

2. Key Features

For Students

- Browse and enroll in courses, track lesson-by-lesson progress
- Attempt quizzes and view scored results
- Earn XP, levels, achievements, and maintain learning streaks (gamification)
- Skill tracking mapped against courses, with AI-suggested learning paths for target job roles
- Career module: job-role readiness matching based on skills acquired
- Downloadable PDF course-completion certificates
- Chat directly with instructors (with attachments and voice notes)
- Age-adaptive themed UI (School / Intermediate / College / University / Other)

For Instructors

- Create and manage courses, lessons, quizzes and questions
- Track student progress, time-on-task, and analytics per course
- Issue, revoke and reissue certificates
- Manage skill mappings and streak rewards
- Two-way chat with enrolled students

For Administrators

- Approve/reject new instructor and student sign-ups (role-based approval workflow)
- Full CRUD over users, courses, lessons, quizzes and questions
- Visual analytics dashboards (custom-built donut/bar charts)
- Excel export of platform activity via openpyxl
- Broadcast announcements to students, instructors, or everyone

Platform-wide

- AI Mentor chatbot (OpenRouter / GPT-4o-mini) offering roadmap, tool and project guidance
- Google OAuth sign-in alongside classic email/username login (django-allauth)
- Light/Dark mode toggle
- Near real-time notification badges and chat via polling

3. System Architecture & Module Breakdown

The application follows Django's Model-View-Template (MVT) pattern. Each app below owns its own models, views, and templates, and is composed together through the root URL configuration and a shared base template.

App	Responsibility
accounts	Custom auth & profile: roles (Admin/Instructor/Student), approval workflow, onboarding, XP/level, presence tracking, notifications.
courses	Core domain model - courses, lessons, quizzes, enrollment, progress, achievements, streaks, skills, learning paths, certificates, instructor-student chat.
students	Student-facing dashboard, catalogue, enrollment, quiz attempts, certificate PDF generation, AI learning-path suggestions.
instructors	Instructor dashboard - lesson/quiz/question CRUD, student analytics, certificate issuance.
adminpanel	Admin dashboard - user/course approvals, CRUD, analytics charts, Excel exports.
mentor_ai	Site-wide AI mentor chatbot logic and conversation handling.
career	Job-role definitions and skill-based career readiness matching.
messaging	Admin-to-user broadcast announcements with read tracking.
utils	Cross-cutting helpers, e.g. Excel-based user activity logging.

4. Technology Stack

Backend

- Python 3 & Django 5.2.9 (MVT architecture, WSGI)
- Classic server-rendered Django views (no DRF/REST layer) with some AJAX/JSON endpoints
- python-dotenv for environment configuration

Database

- MySQL, accessed via PyMySQL (installed as the MySQLdb driver)
- Django ORM for all data modelling and migrations

Frontend

- Server-rendered Django templates (no SPA/build pipeline)
- Bootstrap 5.3.2 + Bootstrap Icons (via CDN)

- Google Fonts (Inter)
- Custom vanilla JavaScript & CSS: theming, dashboard sidebar, chat, broadcast notifications, kids-mode sound/celebration effects
- Hand-built inline SVG/CSS analytics charts (donut & bar) - no external charting library

Authentication & Security

- django-allauth 65.18.0 - Google OAuth social login
- Custom username/email + password authentication in parallel
- Role-based access control (Admin / Instructor / Student) with an approval workflow
- Custom middleware for onboarding completion and last-seen/presence tracking

AI / Intelligence

- OpenRouter API (model: openai/gpt-4o-mini) for the AI Mentor chatbot and AI-generated learning-path suggestions
- Intent detection (roadmap / tools / practice / project / site-help) driving a persona-based mentor prompt

Documents, PDF & Reporting

- ReportLab - programmatic PDF generation for course-completion certificates
- openpyxl - Excel export for admin reporting and activity logs
- Pillow - image processing for uploaded profile, course and lesson images
- xhtml2pdf and pyHanko present in the dependency stack, staged for future HTML-to-PDF templating and digital certificate signing

Messaging

- Database-backed broadcast announcements and instructor-student chat (attachments & voice notes)
- Near real-time UI via client-side polling (fetch + setInterval), no WebSockets/Channels

Deployment & Infrastructure

- Gunicorn WSGI server, Procfile-based deployment (Railway/Heroku-style)
- WhiteNoise for compressed static file serving directly from the app
- Environment-driven configuration for allowed hosts, proxy SSL headers and CSRF trusted origins

5. Database Design

The platform uses MySQL as its relational store, modelled entirely through the Django ORM. Core entities include Users & Profiles (role, approval status, XP/level), Courses, Lessons, Quizzes & Questions, Enrollments & Progress, Skills & Career Job Roles, Achievements & Streaks, Certificates, and messaging entities (Broadcasts, Instructor-Student Chat). Media uploads (profile, course and lesson images, chat attachments, voice notes) are stored on local disk under a structured media/ directory.

6. Authentication & Security

- Sign-in via email/username + password, or Google OAuth (django-allauth)
- Role-based dashboards with a pending-approval gate for new Instructor/Student accounts
- Middleware-enforced onboarding completion before a new user can access the rest of the site
- CSRF protection, HTTPS-aware settings (SECURE_PROXY_SSL_HEADER, CSRF_TRUSTED_ORIGINS) for proxy deployments

7. AI Mentor & Learning-Path Intelligence

A persistent AI Mentor widget is available to logged-in students site-wide. It uses the OpenRouter API (GPT-4o-mini) with a persona-driven system prompt and lightweight intent detection to answer roadmap, tool, practice and project questions, keeping the last few turns of conversation as context. Separately, the AI Learning Path feature recommends the top technologies/tools to learn for a target job role, and stores query history for the student.

8. Deployment & DevOps

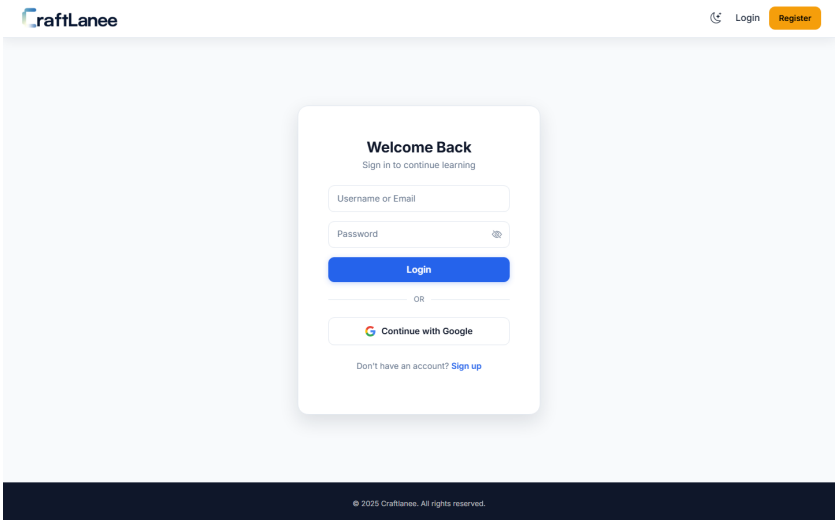
- Procfile runs migrations, collects static files, then starts Gunicorn - a Heroku/Railway-style deployment flow
- WhiteNoise serves compressed, manifest-hashed static assets without a separate CDN
- Environment variables (via python-dotenv) drive secrets, database credentials and allowed hosts across environments

9. Application Screenshots

Craftlanee brand mark, as used across the platform's navigation and login screens:

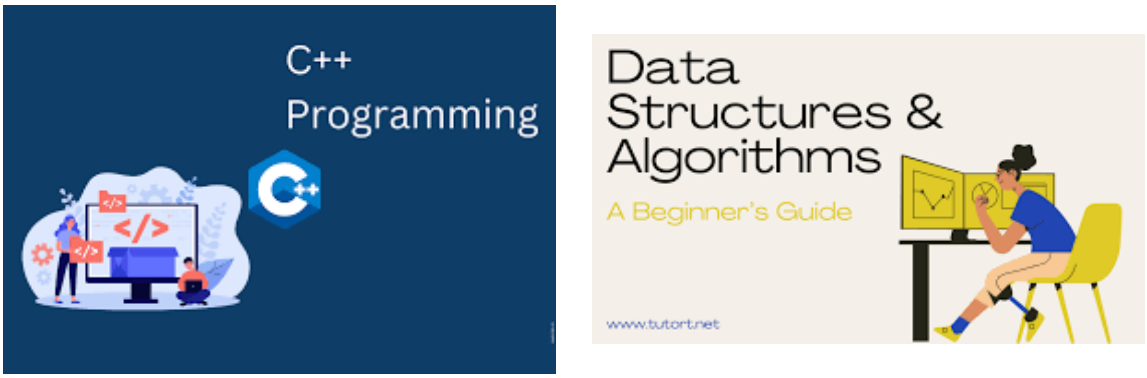
Craftlanee logo

Login screen - Craftlanee E-Learning Platform:



Login page

Sample course cards from the course catalogue:



Course catalogue cards

10. Future Enhancements

- Wire up xhtml2pdf/pyHanko for templated, digitally-signed certificates
- Introduce a REST API layer (Django REST Framework) for a future mobile app
- Move real-time chat/notifications from polling to WebSockets (Django Channels)
- Add automated tests and CI/CD pipeline (none currently in place)
- Move media storage to cloud object storage (e.g. S3) for scalability

11. Conclusion

The Craftlanee E-Learning Platform demonstrates a complete, production-oriented LMS built on a pragmatic Django stack - combining role-based learning workflows, gamification, AI-assisted mentorship, and PDF certification, all deployed through a lightweight, cost-effective Gunicorn + WhiteNoise + Railway/Heroku-style pipeline.